

Øut of Order

The Empty Set and Intervention in Featural Relativised Minimality

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Introduction

This technical note proposes a partial revision of the formalisation of Relativised Minimality (RM; Rizzi 1990, 2004, 2018), arguably the most well-known account of intervention locality effects. RM, as currently defined, draws on set-theoretic relations: Identity, Inclusion, Intersection, and Disjunction. While all are typically used in presentations of the theory, we argue that Disjunction should be excluded, as it entails a formal contradiction. We propose instead that disjunction cases fall outside the scope of RM-computations, as the relevant structural configurations lack an intervener altogether. This move also avoids certain set-theoretical issues involving the empty set.

Featural Relativised Minimality

Featural RM (fRM) posits that movement from Y to X is blocked if an element Z: (i) c-commands Y but not X (intervention condition); and (ii) shares “Relevant Syntactic Features” (RSF) with X (similarity condition), where RSF are features licensing applications of internal and external merge (Rizzi 2018: 347). Crucially, similarity is gradient rather than binary: X and Y are compared via a set-theoretic computation within the Distinctness Hierarchy, which (minimally) defines relations of identity, inclusion, and disjunction. An evaluation procedure at the interface assigns a different penalty to each of these relations. Configurations where the target and the intervener are identical are assigned the strongest penalty. A weaker

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penalty is associated with inclusion, and no penalty is assigned with disjunction. Rizzi (2018) illustrates this pattern in (1), where the degree of deviance is a function of the set-relation at stake and its associated penalty: (1a) is more deviant than (1b) due to identity vs. inclusion, while (1c) is unproblematic as it involves disjunction.

- | | | | |
|-----|----|---|--------------------|
| (1) | a. | * What do you wonder who could buy __ ? | Identity |
| | | [+Q] [+Q] | |
| | b. | ?? What book do you wonder who could buy __ ? | Inclusion |
| | | [+Q; +N] [+Q] | |
| | c. | What do you think he could buy __ ? | Disjunction |
| | | [+Q] [] | |

(Adapted from Rizzi 2018: 348)

Disjunction and intervention

In (1c), the subject *he* is represented as an intervener devoid of RSF. We argue that this is where the contradiction arises. Disjunction is defined as the relation holding when “no feature in the intervener is included in the target” (Belletti et al. 2012: 1064). However, a proper intervener must structurally intervene via c-command and “match [the target] X in terms of RSF” (Rizzi 2018: 347). An element cannot, therefore, simultaneously be an intervener and stand in a disjunction relation with the target.

As a consequence of this contradiction, disjunction cases cannot possibly enter RM-computations. These are conceived as procedures operating at the representational level (Rizzi 2013) over configurations involving two elements, namely a target and an intervener. We argue that cases like (1c) are invisible to this procedure, as no element between *what* and its gap qualifies as a proper intervener under the definition above.

To clarify, this reconceptualisation avoids the contradiction related to disjunction while preserving the (correct) prediction regarding the grammaticality of cases like (1c). Another welcome consequence concerns further problematic implications of the set-theoretic relations discussed in fRM, which emerge once the formal details are worked out.

The problem of the empty set

Let us assume, for the sake of argument, that—contrary to our proposal—disjunction is indeed a valid relation for RM-computations. This would allow for interveners that share no RSF with the target. From the perspective of fRM evaluation,

then, the set of features of the intervener is empty. Consider (1c): while *he* is morphosyntactically specified, it contributes no relevant features in the context of A'-movement, which is the dependency undergoing RM-computation in this example.

Yet assigning no RSF to an intervener entails a non-trivial set-theoretical issue.

The empty set is a subset of any set

By definition, a set B is a subset of a set A if and only if:

$$\forall x (x \in B \rightarrow x \in A).$$

Setting $B = \emptyset$, we must show that:

$$\forall x (x \in \emptyset \rightarrow x \in A).$$

Since $\emptyset$ has no elements, the statement $x \in \emptyset$ is always false. In logic, an implication $P \rightarrow Q$ is vacuously true when P is false. Thus, the statement holds for all x , and we conclude:

$$\forall A (\emptyset \subseteq A).$$

□

The empty set is disjoint from any set

By definition, two sets A and B are disjoint if and only if their intersection is empty:

$$A \cap B = \emptyset.$$

Setting $B = \emptyset$, we compute:

$$A \cap \emptyset = \{x \mid x \in A \text{ and } x \in \emptyset\}.$$

Since $\emptyset$ has no elements, the condition $x \in \emptyset$ is always false. Therefore, no element satisfies $x \in A$ and $x \in \emptyset$, meaning:

$$\forall A (A \cap \emptyset = \emptyset).$$

□

Thus, given A and $\emptyset$, the relation between the two is both disjointness (in RM terms, disjunction) *and* inclusion, formally:

$$\forall A (\emptyset \subset A) \wedge (A \cap \emptyset = \emptyset)$$

While this does not constitute a mathematical contradiction (Halmos 1974), it poses a challenge in a system where these two relations are expected to yield contrasting predictions. In (1c) (repeated in (2)), for instance, the RM-computation between the featural bundles of the target (*what*) and the intervener (*he*) permits both inclusion and disjunction as possible outcomes. This is problematic, since inclusion is expected to yield a (mild) penalty, whereas disjunction involves no penalty, correctly accounting for the absence of intervention effects.

- (2) What do you think he could buy __ ? **Disjunction**
[+Q] []

Crucially, this issue generalises to all cases where an intervener devoid of RSF may yield conflicting set-theoretic relations, leading to inconsistencies in the system’s predictions. Note, moreover, that if Disjunction is not excluded from the set of RM-relevant relations, then any element that merely satisfies the configurational c-command requirement qualifies as an intervener. This is because treating disjunction as a relevant relation allows the system to count elements as interveners even when they do not meet the similarity condition stated above. In this scenario, such ambiguity regarding the intervener-status of Z in [X [Z [Y]]] configurations could apply to any element that c-commands Y and not X (e.g. even the verb *think* in 1c).

Consequences for modelling

Our reconceptualisation has a direct impact on the implementation of RM in two domains. The first is the computational modelling of formal grammatical theories, in the spirit of, e.g., Ozaki and Oseki (2023) and Ginsburg and Fong (2019). In light of the reasoning above, the traditional understanding of RM and our revision imply distinct algorithms for executing RM-computations, differing crucially in the number of comparisons drawn and the corresponding steps required.

The second domain is the statistical modelling of intervention effects. Including Disjunction as a level of a nominal predictor like “Degree of similarity”—alongside Identity or Inclusion (as in, e.g., Friedmann et al. 2009)—treats an arbitrarily selected element as a disjunction intervener. As mentioned, in (1c) there is no principled reason to select *he*, rather than another element satisfying the relevant configurational requirement, such as *think*. Under our proposal, the configuration should instead be treated as a non-intervention baseline, without designating any element as the intervener. Statistically, the presence versus absence of an intervener must therefore be distinguished from the degree of similarity between a target and

a proper intervener. Comparisons between non-intervention configurations and Identity or Inclusion estimate the effect of the presence of an intervener, whereas the comparison between Identity and Inclusion estimates the effect of different degrees of similarity among proper interveners.

Conclusion

To summarise, our note addresses a logical tension within RM. We propose a revision that not only ensures the formal soundness of the theory but also allows for a more precise application of RM in computational and statistical modelling.

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